

Kalyani Patrikar

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Building multiscale computational frameworks with quantum chemical methods, physics-based models, and neural networks, to uncover mechanisms, enhance system performance, and design new materials

EXPERIENCE

Postdoctoral Research Associate, *University of Liverpool*

PRESENT

- Developed probabilistic model integrating molecular dynamics data and charge transport simulations for materials with time variant disorder; **exceeding** state-of-the-art mobility prediction **accuracy** for OMIECs
- Developing **algorithm** simulating charge transport in polymers based on asymmetric charge transfer rates

Early Career Fellow, *Indian Institute of Technology Gandhinagar*

Aug 2022 – Oct 2024

One of the highest postdoctoral level fellowships offered by any Indian Institute

- Processed SMILES and molecular properties into one hot encodings, developed **graph neural network** based surrogates with high prediction accuracy for ground state and excited state properties
- Designed and built **automated** computational pipeline processing molecular SMILES into classified **substructures** for generative design algorithm, 500 000 **novel molecules** analysed by MPNN
- Developed a **multi-scale** framework (ab initio computation, semi empirical QM, kinetic Monte Carlo) to predict system-level behaviour based on **molecular structures**, validated against experimental parameters
- Fine tuned MACE-OFF **MLIP** model (sampling, labelling, training, validation) on AQx-based configurations to enable rapid conformer prediction across the AQx molecular family
- Developed model based on DFT computations and electron transfer theory for mechanism at heterogeneous interfaces predicting observed behaviour of **self assembled monolayers** in devices and catalysts

Senior Research Assistant, *Center for Excellence in Nanoelectronics*

Jul 2013– Dec 2014

- Designed and fabricated oxide nanoparticles embedded organic transistors for toxic gas sensors

SKILLS

Python Libraries : PyTorch (&PyG), RDKit, MACE, ASE, PySCF, NumPy, SciPy, Pandas

Software : CP2K, Gaussian16, Quantum Espresso, xTB, Jmol, GROMACS

AI Models : Graph Neural Networks (GAT, MPNN), CNN, Random Forest

Cheminformatics : substructure matching, molecular fingerprints, molecular graphs, classification, clustering, dimensionality reduction, sampling, visualisation, diversity analysis

KEY PUBLICATIONS (COMPLETE LIST AT kalyanipatrikar.github.io/publications)

Journal (*Equal Contribution Authors, † Corresponding Author)

- Dinesh Trivedi*, Kalyani Patrikar*, Anirban Mondal, "Dataset Size Invariant Graph Attention Network for Prediction of Diverse Molecular Properties", *Mol. Syst. Des. Eng.*, 9, 2024 [10.1039/D4ME00113C]
〈 **Versatile graph neural network surrogates for expensive molecular property prediction** 〉
- Bibhas Das, Kalyani Patrikar, Atharva Keny, Anirban Mondal, "From fragments to function: data-driven design of high-performance non-fullerene acceptors for organic photovoltaics" *Mol. Syst. Des. Eng.*, 11, 269-291, 2026 [10.1039/D5ME00182J] 〈 **End-to-end computational pipeline for combinatorial generative design of high-performance materials** 〉

3. Kalyani Patrikar, Keval Patadia, Rudranarayan Khatua, Anirban Mondal, "From Molecules to Devices: A Multiscale Approach to Evaluating Organic Photovoltaics" *J. Chem. Theory Comput.*, 20, 22, 2024 [[10.1021/acs.jctc.4c01029](https://doi.org/10.1021/acs.jctc.4c01029)] < **Multiscale simulation pipeline DFT → xTB → kMC mapping molecular structure to system behaviour** >
4. Kalyani Patrikar[†], Anirban Mondal[†], "Deciphering the Microscopic Phenomenon Behind Contact Resistances in Interlayer Functionalized Electrodes and Organic Semiconductors", *Physical Review Materials*, 8, 054606, 2024 [[10.1103/PhysRevMaterials.8.054606](https://doi.org/10.1103/PhysRevMaterials.8.054606)] < **First principles model demonstrating mechanisms at heterogenous functionalised interfaces** >

Conference (*Presenting Author)

1. K. Patrikar*, C. Burke, A. Troisi, "Charge Transport Enhancement with Dynamic Disorder", Oral presentation at *European Conference in Molecular Electronics 2025* < **Probabilistic model for dynamically disordered polymers** > Awarded **Best Contributed Talk**

EDUCATION

Doctor of Philosophy

Indian Institute of Technology Bombay

Electrical Engineering

Jan 2015 – Dec 2021

Master of Technology

Indian Institute of Technology Bombay

Materials Science

Jul 2011 – Jun 2013

Bachelor of Technology

Visvesvaraya National Institute of Technology

Metallurgical Engineering & Materials Science

Jul 2007 – May 2011

PHD THESIS

Bulk and Interfacial Charge Transport in Organic Field Effect Transistors

- Demonstrated **novel** charge doping **mechanism** of **self-assembled monolayers** at transistor contacts based on **ab initio computations**, physics-based model, and device characterisation
- Established new **molecular design** paradigm for order-of-magnitude device performance improvement based on device characterisation, **DFT computations**, and **polaron transport** analysis of DA polymers
- Developed **multiscale model** simulating measured cryogenic device characteristics of organic transistors

LEADERSHIP AND COMMUNICATION

- **Teaching:** Flexible Electronics (2 Credit Course) *IIT Gandhinagar*
- **Demonstrator:** Introduction to Spectroscopy, **Chemical Data Discovery and Design** *University of Liverpool*
- **Teaching Assistant:** Machine Learning, Thermodynamics, **Communication Skills** *IIT Bombay*
- **Students Mentored:** BTech and MSc. students for thesis project work *IIT Gandhinagar*
- **Reviewer:** *Applied Physics Letters, Electron Device Letters, etc.*
- **Extracurricular Achievements**
 - Selected as department representative in hosting international delegation of industrialists
 - Organised and anchored events as the Institute Executive Member in IIT Bombay Academic Council